MILITARY SPECIFICATION

ELECTRON TUBE, IMAGE CONVERTER (INFRARED SENSITIVE), TYPE 8598

1. SCOPE

1.1 This specification covers the type 8598 infrared sensitive image converter electron tube.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-N-290	- Nickel Plating (Electrodeposited)
PPP-B-566	- Box, Folding, Paper-Board
PPP-B-601	- Box, Wood, Cleated-Plywood
PPP-B-621	- Box, Wood, Nailed and Lock-Corner
PPP-B-636	- Box, Fiberboard
PPP-C-843	- Cushioning Material, Cellulosic

Military

MIL-P-116	- Preservation, Methods of

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U. S. Military Property
MIL-STD-781	- Reliability Tests:Exponential Distribution

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

JOINT ELECTRON DEVICE ENGINEERING COUNCIL - Color Table

(Application for copies should be addressed to Electronics Industry Association, Engineering Department, 2001 Eye Street, N.W., Washington, D.C. 20006.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to Electronics Inudstry Association Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 Description. - The image converter electron tube shall be of the monovoltage 8598 type having an electrostatic focus; an infrared sensitive (S-1 spectral response) photocathode; an aluminized P-20, phosphor-image screen; a minimum conversion index of 10; an operating potential of 12,000 volts; and a useful cathode diameter of 0.625 inch.

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3.2 Preproduction. - The supplier shall furnish tubes (see 6.2) within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods will produce tubes that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish tubes conforming to this specification.

3.3 Initial production. - When specified the supplier shall furnish to the Government tubes (see 6.2) for inspection as specified in 4.4.

3.4 Material. - Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Photocathode. - The photoemissive material of the photocathode shall have an S-1 spectral response.

3.4.2 Image screen. - The sensitive material of the image-forming screen shall be type P-20 aluminized phosphor, with a special energy emission characteristic which peaks at 5,300 angstroms plus or minus 100 angstroms in the standard color table (JEDEC). The color coordinates shall be controlled within the following limits:

X = 250	Y = 600
X = 350	Y = 500

3.5 Dimensions and weight.

3.5.1 Dimensions. - The tube shall be constructed in accordance with Figure 1 and as specified herein.

3.5.2 Weight. - The weight of the tube shall not exceed one (1) ounce.

3.6 Operation and environmental.

3.6.1 Conversion index. - The image converter tube shall have a conversion index (CI) of not less than 10.

3.6.2 Conversion-index decay. - After a holding period of 672 hours (4 weeks) which shall not include the manufacturers stabilizing period of at least 336 hours (2 weeks), the conversion index of the tube shall be not less than 10 and shall have slumped not more than 20 percent from the initial value measured at the end of the stabilizing period. When the conversion index has slumped more than 20 percent but is still above the minimum value of 10, the tube shall be held an additional two weeks. At the end of the 6 week period, the tube shall have a conversion index of not less than 10 and shall have slumped not more than 30 percent of the initial value. Rejuvenation of photocathodes by any means is prohibited.

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3.6.3 Conversion index to background. - The conversion index (CI) to background brightness (B) ratio shall be not less than 4.5.

3.6.4 Irradiation surge. - The image of a source on the screen of the tube shall not disintegrate nor shall any part of the image darken or be blanked out when a minimum of 2.0 microamperes of photocathode current is developed by exposing the photocathode to radiation from a tungsten source for 5 seconds.

3.6.5 Image alinement. - The center of an image produced on the screen of the tube by focusing a test spot on the center of the photocathode shall fall within a 0.03 inch radius circle concentric with the optical axis of the tube.

3.6.6 Image shift. - The center of the test-spot image shall not be displaced more than 0.0025 inch in any direction from its initial position after applying a series of 10 electrical shocks. Each shock shall consist of applying the rated voltage of the tube for 10 seconds and then removing it for 15 seconds.

3.6.7 Center resolution. - The tube shall have a minimum resolution of 64 line pairs per millimeter (LP/MM) within a 4 millimeter circle concentric with the optical axis of the tube. The minimum peak resolution shall be 72 LP/MM and shall occur between 2 circles 2.5 millimeters and 4 millimeters in diameter. There shall be no greater than 10 LP/MM difference between minimum center resolution and minimum peak resolution.

3.6.8 Peripheral resolution. - The tube shall have a minimum peripheral resolution of 12 LP/MM at a 5 millimeter radius from the optical center.

3.6.9 Center magnification. - The center magnification of the tube shall be not less than 0.95 nor greater than 1.05. The center magnification shall be measured with a nominal chordal separation of object points of 20 percent plus or minus 0.4 percent of the minimum useful diameter of the photocathode.

3.6.10 Distortion. - The tube shall introduce image distortion of not more than 14 percent as determined by:

$$D = \frac{M_2 - M_1}{M_1} \times 100$$

WHERE:

D = Percent distortion

M_1 = Magnification determined in center magnification test

M_2 = Magnification measured with a nominal chordal separation of object points of 55 percent plus or minus 0.4 percent of the minimum useful diameter of the photocathode (0.625 inch) (see figure 1).

3.6.11 Cathode and screen spots. - When the screen of the tube is viewed with a 2.5 power magnifier and with no radiation incident on the photocathode, there shall be no ion spots (see 6.3.1) or cold emission (see 6.3.2) that appear as bright streaks or blemishes. When the screen is viewed through a 10-power micro-

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scope and with the radiation level on the photocathode adjusted to obtain best spot contrast, the opaque or dark spots (see 6.3.3) shall not exceed the size and quantities specified in Table I. Size of noncircular spots shall be determined on the basis of equal area to circular spots. When the distance between two spots is less than the maximum dimension of either spot, the two spots shall be considered as one spot with a size equal to the sum of the maximum dimensions of the two spots plus the amount of separation between them.

TABLE I SIZE OF SPOTS

Maximum diameter of spots (inches)	Number of spots within 6 mm diameter circle	Number of spots within area bounded by two circles 6mm and 10 mm in diameter
0.001 to 0.003	5	8
0.003 to 0.006	1	4
0.006 to 0.012	0	2

NOTE: The 6 and 10 mm circles on the image screen shall be concentric with the optical axis of the tube.

3.6.12 Cathode and screen uniformity. - The background shading of the screen of the tube shall be uniformly graded with no line of demarcation between light and dark areas under the following conditions.

a. With no radiation incident on the photocathode.

b. With radiation from a tungsten source of 0.1 to 0.4 lumen per square foot, passed through a Corning No. 2540 filter (melt No. 1613, of 2.61 mm thickness), or equivalent filter, uniformly distributed over the photocathode.

3.6.13 Breakdown. - With no radiation incident on the photocathode of the tube, there shall be no arcing, flashing, flickering, or any other intermittent or continuous failure at the operating potential of 12 kilovolts or when an overvoltage of 1 kilovolt (13 kilovolts total) is applied for 10 seconds.

3.6.14 Dark current. - The total dark current (see 6.3.4) of the tube shall not exceed 0.02 microampere when measured in an atmospheric environment of 40 to 50 percent humidity and with no radiation incident on the photocathode.

3.6.15 Vibration. - The tube shall not be damaged (see 6.3.5) and shall meet all operational requirements after being rigidly mounted, singly, or in groups, and

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vibrated with simple harmonic motion parallel and perpendicular to the optical axis of the tube over a frequency range of 10 to 55 cycles per second (cps) at an amplitude of not less than 0.125 inch total excursion for 10 minutes in each direction with no voltage applied during the vibration.

3.6.16 Mechanical shock. - The tube shall operate without flashing or electrical breakdown during more than two shock pulses in each direction while being subjected to six 75-g (see 6.3.6) shocks applied parallel to the longitudinal axis and six 75 g-shocks applied perpendicular to the longitudinal axis of the tube.

3.6.17 Storage temperature.

3.6.17.1 High temperature storage. - The tubes shall not be damaged by storage in any ambient temperature up to plus 68° C with maximum humidity possible throughout the temperature range.

3.6.17.2 Low temperature storage. - The tubes shall not be damaged by storage in any ambient temperature to minus 54° C with maximum humidity possible throughout the temperature range.

3.6.18 Operational temperature.

3.6.18.1 High temperature operation. - The tube shall perform as specified herein at any ambient temperature to plus 52° C at all possible relative humidities within that range.

3.6.18.2 Low temperature operation. - The tube shall perform as specified herein at any ambient temperature to minus 32° C with all possible relative humidities within that range.

3.6.19 Thermal shock. - The tubes shall not be damaged by temperature shocks from plus 52° C to room temperature (see 6.3.7) and from minus 54° C to room temperature.

3.6.20 Mean time between failure. - The tube shall have a specified mean time between failure of 2000 hours.

3.7 Identification marking. - Each tube shall be identified in accordance with MIL-STD-130. The marking shall include a coded acceptance date. Reading from left to right (or top to bottom), the code number shall designate the year and week of acceptance in that order. The first two digits of the code shall be the last two digits of the number of the year. The second two digits of the code shall be two digits indicating the calendar week of the year (01 through 52).

3.8 Workmanship. - All parts and components of the tube shall be free of dirt and other extraneous material and from defects of workmanship that could impair the performance of the tube.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection

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requirements as specified herein. Except as otherwise specified by the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (see 4.3).

b. Initial production inspection (see 4.4).

c . Quality conformance inspection (see 4.5).

d. Inspection comparison (see 4.7).

e. Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. - Each preproduction tube shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

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101. Any dimension not as specified.

102. Any process treatment not as specified.

103. Broken optical or mechanical components.

104. Missing or illegible markings.

105. Workmanship not as specified.

4.3.2 Tests. - Following successful completion of the examination, as specified in 4.3.1, the preproduction tubes shall be subjected to the tests in Table II. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

TABLE II TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
High temperature storage	3.6.17.1	4.6.17.1
Low temperature storage	3.6.17.2	4.6.17.2
High temperature operation	3.6.18.1	4.6.18.1
Low temperature operation	3.6.18.2	4.6.18.2

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TABLE II TEST SCHEDULE (Cont)

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Thermal shock	3.6.19	4.6.19
Vibration	3.6.15	4.6.15
Mechanical shock	3.6.16	4.6.16
Conversion index	3.6.1	4.6.1
Conversion index decay	3.6.2	4.6.2
Conversion index to background	3.6.3	4.6.3
Irradiation surge	3.6.4	4.6.4
Image alinement	3.6.5	4.6.5
Image shift	3.6.6	4.6.6
Center resolution	3.6.7	4.6.7
Peripheral resolution	3.6.8	4.6.8
Center magnification	3.6.9	4.6.9
Distortion	3.6.10	4.6.10
Cathode and screen spots	3.6.11	4.6.11
Cathode and screen uniformity	3.6.12	4.6.12
Breakdown	3.6.13	4.6.13
Dark current	3.6.14	4.6.14
Mean time between failure	3.6.20	4.6.20

4.4 Initial production inspection. - When specified (see 3.2), initial production tubes will be selected at random by the Government from the first tubes produced from production tooling, shall be examined as specified in 4.3.1 and tested as specified in Table II to determine conformance to the requirements of this specification. The inspection will be performed by the contractor under supervision of a representative of the contracting officer. Acceptance of initial production tubes shall not exclude the remaining tubes from the quality conformance inspection and acceptance provisions specified in section 4.

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4.4.1 Inspection failure. - Failure of the initial production tubes to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production tubes and shall be cause for refusal by the Government to continue acceptance of production tubes until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on tubes previously produced and accepted under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all tubes accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance.

4.5.1 Examination. - Each tube shall be examined as specified in 4.3.1. One or more defects shall be cause for rejection of that tube.

4.5.2 Tests.

4.5.2.1 Group A. - Each tube shall be tested in accordance with Table III. Failure of any test shall be cause for rejection of that tube.

TABLE III GROUP A TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Mechanical shock	3.6.16	4.6.16
Conversion index	3.6.1	4.6.1
Conversion index decay	3.6.2	4.6.2
Conversion index to background	3.6.3	4.6.3
Center resolution	3.6.7	4.6.7
Peripheral resolution	3.6.8	4.6.8
Cathode and screen spots	3.6.11	4.6.11

4.5.2.2 Group B. - Group B tests shall be conducted on tubes selected from units which have passed the examination and tests in 4.5.1 and 4.5.2.1. Sample size shall be in accordance with MIL-STD-105, AQL 1.5 percent defective inspection level S-3. Tests shall be conducted in accordance with Table IV and may be performed in any sequence.

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TABLE IV GROUP B TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Irradiation surge	3.6.4	4.6.4
Image alinement	3.6.5	4.6.5
Image shift	3.6.6	4.6.6
Center magnification	3.6.9	4.6.9
Distortion	3.6.10	4.6.10
Cathode and screen uniformity	3.6.12	4.6.12

4.5.2.3 Group C. - This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.2.3.1. The tests shall be performed in the order listed.

4.5.2.3.1 Sampling for inspection. - Ten (10) sample tubes shall be selected by the Government for all inspections in Table V from the monthly specified quantity or every 30 days whichever occurs first. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE V GROUP C TEST SCHEDULE

TEST	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
High temperature operation	3.6.18.1	4.6.18.1
Low temperature operation	3.6.18.2	4.6.18.2
Vibration	3.6.15	4.6.15
Mean time between failure	3.6.20	4.6.20

4.5.2.3.2 Special sampling lot failure. - In the event of failure of the special sampling lot, the contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on

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(i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspection which the sample failed, at the option of the Government) and Groups and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until Group C reinspection results have shown that the corrective action was effective.

4.5.3 Reinspection of conforming group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed both group B and group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Test conditions. - Unless otherwise specified in the test procedures, the radiation source used in the tests shall be a tungsten filament lamp operated at a color temperature of 2870° K; the filter shall be a Corning No. 2540 filter (melt No. 1613, 2.61 mm thickness), or equivalent; and the applied potential shall be 12,000 volts. The filter shall be placed between the radiation source and the photocathode of the tube. The amount of radiation from the source incident on the photocathode for individual tests shall be the amount specified in the test paragraph. The tube shall be subjected to not more than 2.0 microamperes of dc current for a single application of not more than 2 minutes and to not more than 10 such applications. Following the shock test, each tube shall be held for a stabilization period of 336 hours (2 weeks) prior to performing the balance of the tests. Unless otherwise specified all tests shall be performed at room temperature. Chambers used for temperature tests shall maintain the specified temperature within plus or minus 2° C. For thermal shock, the rate of change of ambient temperature shall not be less than 60° C per minute.

4.6.1 Conversion index. - The radiation source and filter specified in 4.6 shall be used to produce a 0.5 inch diameter circular image on the center of the photocathode. The amount of incident flux (F_2) on the photocathode (prior to interposing) the filter) shall be 0.01 lumen plus or minus 5 percent. The luminous flux (F_1) output of the tube image screen shall be read with a Weston Model 856 RRV Photronic Cell or equal provided with a Visco or equal filter and a truncated lucite cone. The base of the cone shall be cemented to the photronic cell window; the top of the cone shall be placed in contact with the image tube to cover the excited area of the image screen. The photronic cell and cone shall be precalibrated from a diffused light source. A conversion index of less than 10 shall constitute failure of this test. The conversion index (CI) is:

$$CI = \frac{F_1}{F_2 t}$$

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WHERE:

F_1 is the total available luminous flux emitted by the phosphor of the image screen of the tube with the operating voltage specified herein applied.

F_2 is the unfiltered flux from the 2870° K light source (see 4.6) incident on the photocathode of the tube.

t is the filter factor (see 6.3.8) of the interposed infrared filter (see 4.6).

4.6.2 Conversion index decay. - The conversion index decay test will be made in the same way and with the same light source and filter as the conversion-index test (see 4.6.1). Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.3 Conversion index to background. - The conversion index (CI) to background brightness (B) ratio shall be made using the same light source and filter with the same amount of incident flux on the photocathode, specified in 4.6.1. With the specified flux incident on the photocathode, a 5819 or IP21 photomultiplier tube (or equivalent) shall be placed with its cathode against the screen faceplate of the image tube. The bleeder voltage of the photomultiplier shall be adjusted to give 220 microamperes of anode current. The flux from the radiation source shall then be excluded from the photocathode and the background brightness shall be read in terms of anode current of the photomultiplier tube. The conversion index to background brightness ratio is (CI/B) = 1.00/LB. A conversion index to background brightness ratio of less than 4.5 shall constitute failure of this test.

4.6.4 Irradiation surge. - The tubes shall be subjected to an irradiation surge by exposing the photocathode to a tungsten light source(which need not be the one specified in 4.6) for 5 seconds to produce a minimum of 2.0 microamperes of photocathode current. Nonconformance to 3.6.4 shall constitute failure of this test.

4.6.5 Image alinement. - A 10-power (or higher) microscope shall be alined with the optical axis of the tube. With the microscope focused on the screen of the tube and with the tube at operating potential, a test spot shall be projected on the center of the photocathode and image of the spot shall be observed on the screen. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.6 Image shift. - Following the alinement test, the position of the center of the test spot shall be recorded. A series of 10 electrical shocks shall then be applied to the tube. Each shock shall consist of applying the operating potential for 10 seconds and then removing it for 15 seconds. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.7 Center resolution. - The resolution test shall be performed using:

a. A 10 power or higher microscope.

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b. A tungsten radiation source and infrared filter which need not be the source and filter specified in 4.6.

c. A condenser lens system to project the resolution test pattern at unity magnification upon the photocathode of the image tube. (Projection other than unity magnification may be used provided the chart size is such that the projected image on the photocathode meets the requirements of figure 2).

The test pattern shall be placed between the radiation source and the filter and shall be focused on the photocathode for best resolution of line-sets in the central area of the screen of the image tube. The input radiation level shall be adjusted to obtain the greatest contrast of the resolution pattern. The image is resolved when all lines in each pattern (see groups marked 64 and 72 in figure 2) can be counted. Failure of the tube to resolve the lines shall constitute failure of this test.

4.6.8 Peripheral resolution. - The test procedure for conducting this test shall be that specified in 4.6.7. The test patterns (two groups) shall be those marked 45 and 12 on figure 2. The peripheral images are resolved when all lines in each resolution group can be counted. Failure to resolve the lines shall constitute failure of this test.

4.6.9 Center magnification. - The center magnification shall be measured with the object points projected on the photocathode with nominal monochromatic visible light. The nominal chordal separation of the object points shall be 20 percent plus or minus 0.4 percent of the useful minimum cathode diameter. Nonconformance to 3.6.9 shall constitute failure of this test.

4.6.10 Distortion. - The percent distortion of the tube shall be obtained from a second magnification measurement. The test procedure shall be the same as for the center magnification except for the spacing of the object points. The object points projected on the photocathode shall have a separation of 55 percent plus or minus 0.4 percent of the useful minimum cathode diameter (see figure 1). The percent distortion shall be obtained by use of the formula specified in 3.6.10. Distortion values not falling within the limits specified in 3.6.10 shall constitute failure of this test.

4.6.11 Cathode and screen spots. - The operating potential shall be applied to the tube, and with no radiation incident on the photocathode, the image screen shall be observed with a 2.5 magnifier for ion spots or cold emission that appear as bright streaks or blemishes. The photocathode shall then be exposed to radiation level adjusted for best spot contrast and the image screen shall be observed with a 10 power (or higher) microscope for opaque or dark spots. Nonconformance to 3.6.11 shall constitute failure of this test.

4.6.12 Cathode and screen uniformity. - With the operating potential applied to the tube, the screen shall be viewed with a 2.5 magnifier in two steps as follows:

a. With no radiation incident on the photocathode.

b. With radiation uniformly distributed over the photocathode.

The radiation level, prior to filtering with the filter specified in 4.6 shall be 0.1 to 0.4 lumen per square foot. Nonconformance to 3.6.12 shall constitute failure of this test.

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4.6.13 Breakdown. - This test shall be conducted with no radiation incident on the photocathode. With the normal operating potential applied, the tube shall be observed for evidence of breakdown. When there is no evidence of breakdown, the potential shall be increased to 13 kilovolts for 10 seconds. Evidence of breakdown as specified in 3.6.13, shall constitute failure of this test.

4.6.14 Dark current. - The dark current shall be measured with a microammeter connected in series with the grounding terminal of the tube. The test shall be performed without radiation incident on the photocathode and in an atmospheric environment of 40 to 50 percent relative humidity. A dark current of more than 0.02 microampere shall constitute failure of this test.

4.6.15 Vibration. - The tubes, rigidly mounted singly or in groups in a vertical position with the cathode end up, shall be subjected to a simple harmonic motion parallel and perpendicular to the optical axis having an amplitude of not less than 0.0625 inch plus or minus 0.0025 inch (0.125 plus or minus 0.005 inch total excursion) at a frequency of 10 Hz to 55 Hz. In one minute, vary the frequency from 10 to 55 Hz and return to 10 Hz. Repeat this frequency sweep 10 times in each direction. No voltage shall be applied to the tube during the test. The tubes shall be examined for any physical damage resulting from the vibration. Evidence of damage shall constitute failure of this test.

4.6.16 Mechanical shock. - The tube, in a test fixture, shall be placed in a darkened room (not more than 0.01 footcandle) and energized with 12,000 volts for a minimum period of 15 seconds. At the end of this period, with the tube still energized and no light incident on the photocathode, the tube shall be subjected to six shock impacts applied parallel to and six shock impacts applied perpendicular to its longitudinal axis. The shock impacts shall be applied in such a way as to generate nominal half sine wave pulses having a minimum of 75 g's at the peak. The duration of each shock pulse shall be 6 milliseconds plus or minus 2 milliseconds measured between the 10 percent values of the peak amplitude. The energy under the shock curve shall be not less than 0.25 g seconds and the after oscillations shall be not greater than 15 percent of the peak amplitude of the nominal half sine wave pulse. Nonconformance to 3.6.16 shall constitute failure of this test.

4.6.17 Storage temperature.

4.6.17.1 High temperature storage. - The tubes shall be exposed to an ambient temperature of plus 68° C with maximum humidity of 90 to 98 percent for a period of 4 hours. At the conclusion of the 4 hour period inspect for damage. Any evidence of damage shall constitute failure of this test.

4.6.17.2 Low temperature storage. - The tubes shall be exposed to an ambient temperature of minus 54° C for a period of 4 hours. At the conclusion of the 4 hour period inspect for damage. Any evidence of damage shall constitute failure of this test.

4.6.18 Operational temperature.

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4.6.18.1 High temperature operation. - Place the tubes in a chamber which has an internal temperature of 52° C and soak for 4 hours. At the conclusion of the 4 hour period, apply operating potential to the tube. Observe the tubes for not less than 10 minutes for electrical failure or malfunctioning. Any evidence of damage shall constitute failure of this test.

4.6.18.2 Low temperature operation. - Place the tubes in a chamber which has an internal temperature of minus 32° C and soak for a period of 4 hours. At the conclusion of the 4 hour period, apply operating potential to the tube. Observe the tubes for not less than 10 minutes for electrical failure, or malfunctioning. Any evidence of damage shall constitute failure of this test.

4.6.19 Thermal shock. - The tubes shall be soaked in an ambient temperature of 52° C for a period of 4 hours. Remove the tubes from the chamber and allow to soak for 2 hours at room temperature. Examine for damage. Place the tubes in a chamber with ambient temperature of minus 54° C and soak for 4 hours. Remove the tubes from the chamber and allow to soak for 2 hours. Examine for damage. Any evidence of damage shall constitute failure of this test.

4.6.20 Mean time between failure. - Mean time between failure testing shall be performed in accordance with MIL-STD-781, Test Level A1, Test Plan VI each 30 days on sample tubes selected at random from the previous 30 day production period. The sample size shall be a minimum quantity of 10. For initial production testing, no replacements are permitted. For quality conformance testing, replacement of failed tubes is required. Sample tubes selected for testing, shall be selected from lots which have passed the quality conformance inspection. Testing shall be conducted in a darkened area. The operating potential shall be applied and there shall be no radiation incident on the photocathode. The tubes shall be operated as specified and the total number of operating hours for all units, failed or unfailed,original and replacement, shall be monitored and recorded. Once each 96 hours the following inspections shall be performed. Time to failure for any tube failing any test shall be computed to the beginning of the 96 hour period during which the failure occurred.

a. Conversion index (4.6.1).

b. Conversion index to background (4.6.3).

c. Cathode and screen spots (4.6.11).

d. Cathode and screen uniformity (4.6.12).

e. Breakdown (4.6.13).

Failure to meet the requirements of 3.6.1, 3.6.3, 3.6.11, 3.6.12, and 3.6.13 shall constitute failure of this test.

4.7 Inspection comparison. - The Government may select tubes at any time during the contract production period and subject these tubes to the examination specified in 4.3.1 and to the tests specified in Table II, to determine that the quality of the selected tubes is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Tubes will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction or initial production tubes.

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4.7.1 Inspection failure. - Failure of any tube when examined and tested as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of tubes until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. - For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Lot. - A lot shall consist of all completed packs offered for examination at one time.

4.8.1.3 Sampling. - Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.4 Examination. - Sample selected in accordance with 4.8.1.3 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Materials and methods not as specified for levels A and B.

106. Tubes not cushioned.

107. Intermediate packaging not as specified for levels A and B.

108. Gross weight exceeds 70 pounds for level A.

109. Strapping not zinc coated with level A.

110. Gross weight exceeds the weight limitation of the box specification for level B.

111. Box not reinforced as specified for level B.

112. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. - Each tube shall be preserved in accordance with MIL-P-116, method IC-1. The tube shall be cushioned on all sides, the top and the bottom with not less than 1-inch-thick wrap of cushioning material conforming to PPP-C-843, Type 1, Class Z, prior to placing the tube in the IC-1 bag. The bag containing the tube shall be placed in a close-fitting box conforming to PPP-B-566, style, type, and class optional.

5.1.1.1 Intermediate packaging. - The tubes shall be intermediate packaged in quantities of 12 in boxes conforming to PPP-B-636, class 2, W6c, style optional.

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5.1.2 Level B. - Each tube shall be cushioned and placed in a box specified in 5.1.1. The method IC-1 bag shall not be required.

5.1.2.1 Intermediate packaging. - The tubes shall be intermediate packaged as specified in 5.1.1.1 except the box shall be type 1, class 1, style optional.

5.1.3 Level C. - Each tube shall be cushioned to prevent breakage and packaged in a close-fitting paperboard box or in the supplier's commercial container in a manner to afford protection from the supplier to the initial destination.

5.2 Packing. - Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. - The tubes shall be packed in quantities not to exceed 70 pounds gross weight in close-fitting boxes conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The straps shall be zinc coated.

5.2.2 Level B. - The tubes shall be packed in a close-fitting box conforming to PPP-B-636, class 2, V3c, or V3s, style optional. The gross weight shall not exceed the weight limitation of the box specification. The box shall be reinforced with bands of reinforced tape in accordance with the appendix to the box specification.

5.2.3 Level C. - The tubes shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in fiberboard containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 Marking. - Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The tube covered by this specification is intended for use in an electro-optical viewing device to convert radiation in the 0.8 to 1.2 micron wavelength band reflected from objects or backgrounds to a visible image.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When preproduction or initial production tests are to be conducted and number of models of either or both which will be needed (see 3.2, 3.3, 4.3 and 4.4).

c. Level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

6.3 Definitions.

6.3.1 Ion spots. - A bright spot on the image screen caused by ionic bombardment with no illumination incident on the photocathode.

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6.3.2 Cold emission spots, streaks and blemishes. - Bright spots, streaks or other configurations of greater intensity than the background brightness of the tube and visible on the image screen with no radiation incident on the photocathode. The imperfections are caused by thermal or field emission.

6.3.3 Dark spots. - Opaque or dark imperfections visible on the image screen of the tube when the photocathode is irradiated.

6.3.4 Dark current. - The electrical current which flows within the tube and across the external surfaces of the tube with no radiation incident on the photocathode.

6.3.5 Damage . - Damage is defined as:

a. Cracks, crazing, breakage or deformation of the envelope or other components of the tube.

b. Electrical failure or malfunctioning, including blanking or flickering of the image screen, or failure to fluoresce.

6.3.6 "g". - g is defined as the acceleration or deceleration of 32.17 feet per second per second caused by the force of gravity.

6.3.7 Room temperature. - Room temperature is defined as 23° C $\pm$ 3°C.

6.3.8 Filter factor. - The filter factor, T, is the percentage of 2870° K holo-lumen flux transmitted by the infrared filter. T is determined by the formula:

$$T = \frac{\int_0^\infty \frac{R\lambda \quad J\lambda \quad t\lambda}{R\lambda \quad J\lambda} \frac{d\lambda}{d\lambda}}{\int_0^\infty}$$

WHERE:

$R\lambda$ = The relative spectral response of an EIA-S1 photo-surface.

$J\lambda$ = The relative spectral distribution of the 2870° K color temperature radiation source.

$t\lambda$ = The spectral transmission of the infrared filter.

$d\lambda$ = The differential of λ.

λ = The wavelength of the radiation.

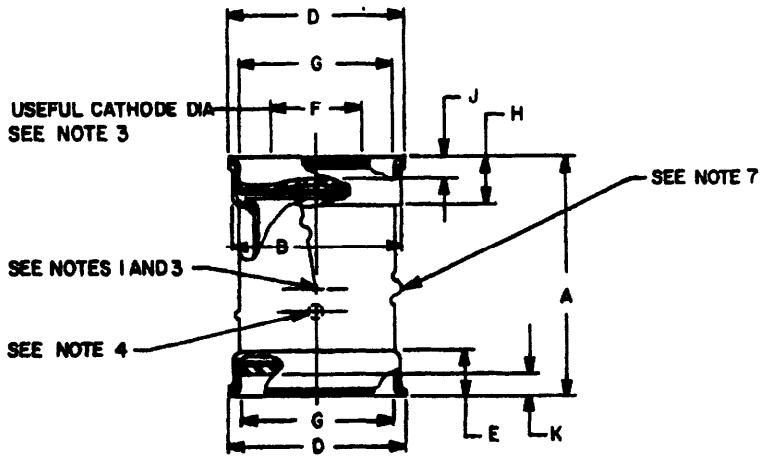
CUSTODIAN:
Army-EL

REVIEWING ACTIVITY:
Army-ME,MU,AV,WC

PREPARING ACTIVITY:
Army-EL

Project Number 5855-A068

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DIM	A	B	D	E	F	G	H	J	K
MAX	1.690	1.107	1.180	.355	----	1.033	.327	.075	.182
MIN	1.627	1.085	1.140	.315	.625	1.025	.287	.043	.150

NOTES:

1. CATHODE FACEPLATE:
   INTERNAL AND EXTERNAL RADIUS OF CURVATURE: 1.220 TO 1.240 INCHES.
   CENTER THICKNESS OF GLASS: 0.055 TO 0.065 INCH.
   MAXIMUM VARIATION IN EDGE THICKNESS: 0.004 INCH.
2. IMAGE SCREEN:
   CENTER THICKNESS OF GLASS: 0.054 TO 0.064 INCH.
   MAXIMUM VARIATION IN EDGE THICKNESS: 0.004 INCH.
3. THE INDEX OF REFRACTION FOR THE CATHODE AND IMAGE SCREEN SHALL BE 1.49 ± 0.04 INCHES.
4. ALL METAL PROCESSING TERMINALS PROTRUDING THROUGH THE GLASS ENVELOPE SHALL BE CUT OFF AND GROUND FLUSH WITH GLASS BEAD. ALL LEADS AND METAL PROCESSING TERMINALS SHALL BE COATED WITH GLYPTOL OR EQUIVALENT AND SHALL NOT EXTEND BEYOND DIAMETER 'B' DIMENSION.
5. THE TUBE AXIS IS DETERMINED BY A STRAIGHT LINE THROUGH THE GEOMETRIC CENTER OF THE DIAMETERS 'G' AT THE TUBE ENDS.
6. THE EXPOSED METAL PARTS OF THE TUBE SHALL BE NICKLE PLATED IN ACCORDANCE WITH QQ-N-290, TYPE III , CLASS 1
7. THE EXHAUST TUBULATION SHALL BE A MINIMUM OF 0.020 INCH LESS THAN DIAMETER 'D' DIMENSION.

FIGURE 1
TUBE, IMAGE CONVERTER 8598

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NOTES:

1. THE RESOLUTION GROUPS SHALL BE HIGH CONTRAST (THE BLACK LINES HAVING A LOG DENSITY OF 2.0 OR MORE) ON A CLEAR BACKGROUND.
2. THE RESOLUTION GROUPS, WHOSE FREQUENCIES ARE INDICATED IN LP/MM, SHALL HAVE 2 SETS OF BAR PATTERNS, 1 SET VERTICAL AND 1 SET HORIZONTAL, EACH OF THE FOUR BLACK LINES 9:1 LENGTH TO WIDTH RATIO. THE INCLUDED CLEAR LINES OF EACH BAR PATTERN ARE TO BE OF THE SAME WIDTH AS THE BLACK LINES.
3. ALL DIMENSIONS ARE IN INCHES.

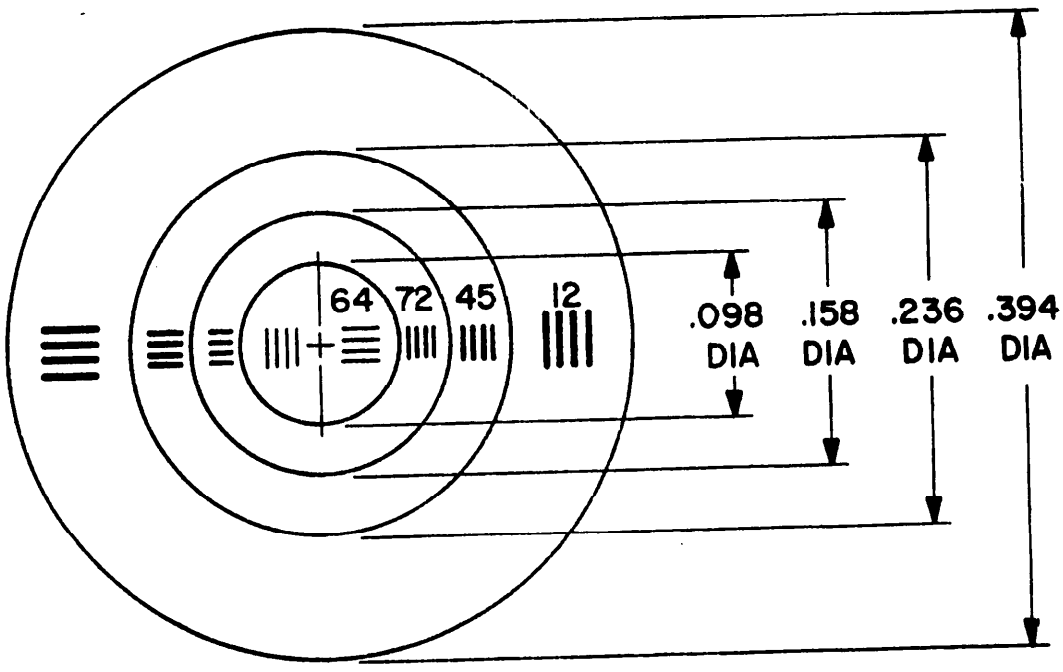


FIGURE 2.
RESOLUTION PATTERN

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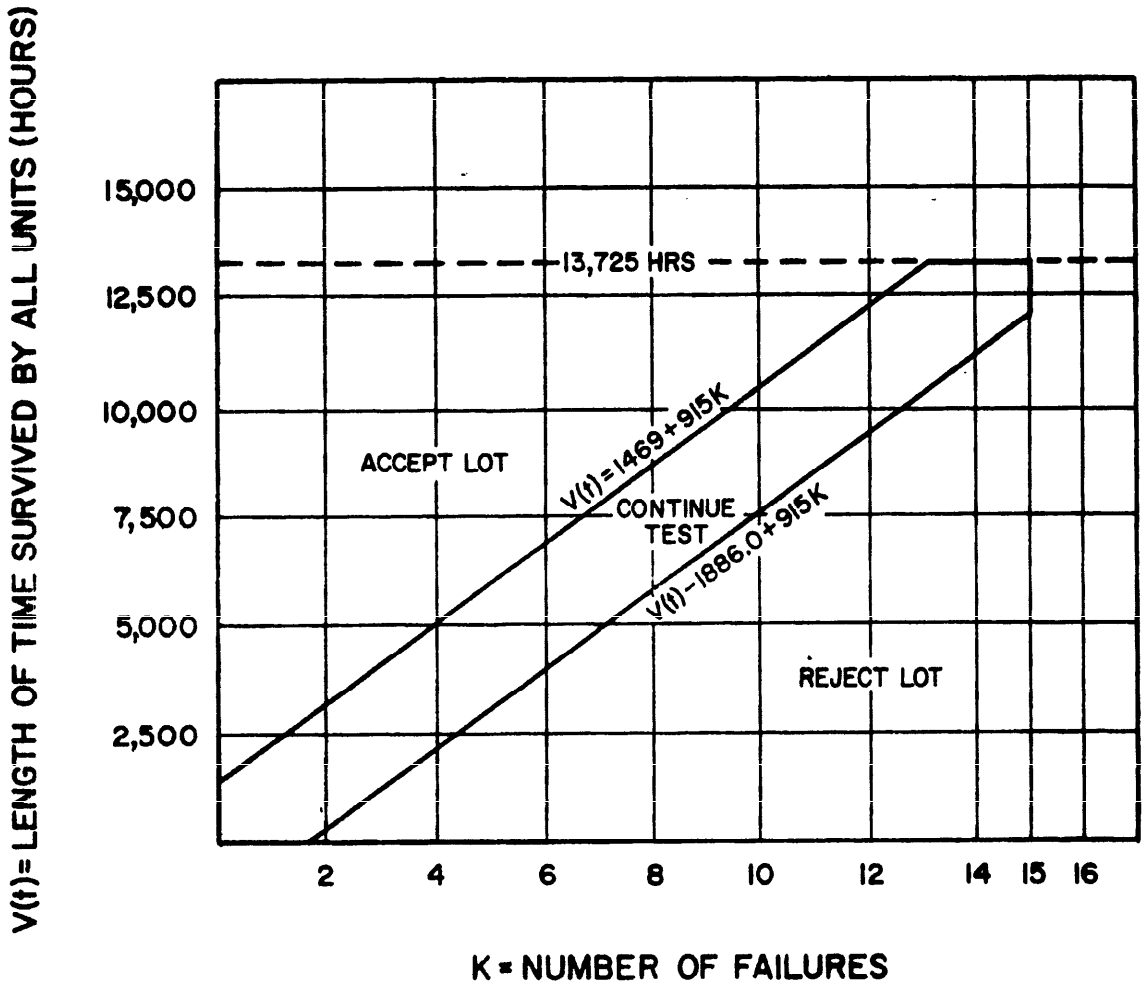


FIGURE 3.
SEQUENTIAL LIFE TEST WITH
REPLACEMENT DECISION GRAPH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-55332(EL), ELECTRON TUBE, LOAD CONVERTER (INFRARED SENSITIVE), TYPE 8598

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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